🛹 Energy Skate Park Simulation Worksheet

Topic: Energy Transformations
Level: National 5 Engineering Science
Simulation Link: https://phet.colorado.edu/sims/html/energy-skate-park/latest/energy-skate-park_all.html

🔍 Learning Objectives

- Understand the relationship between kinetic and potential energy.
- Observe energy conservation in a closed system.
- Investigate the effects of mass, friction, and track design on energy transfer.

🧠 Part 1: Getting Started

Instructions: Open the simulation and select the "Intro" tab.

1. Place the skater on the track. What happens when you release them from the top of the ramp?
   ✎ _____
2. Turn on the **Bar Graphs** and **Pie Chart**. What do these represent?
   ✎ _____
   ✎ _____
   ✎ _____

⚙️ Part 2: Energy Observations

Instructions: Use the simulation to answer the following.

Position on Track	Kinetic Energy	Potential Energy	Total Energy
Top of Ramp			
Middle of Ramp			
Bottom of Ramp			

1. What do you notice about the **total energy** at each point?
   ✎ _____
2. How does the **skater's speed** relate to their kinetic energy?
   ✎ _____
   ✎ _____

🧪 Part 3: Experimenting with Variables

Instructions: Try changing the skater's **mass** and **friction** settings.

1. Increase the skater's mass. What effect does this have on:
   - Potential energy? ✎ _____
     _____
   - Kinetic energy? ✎ _____
     _____
2. Add friction to the track. What happens to the total energy over time?
   ✎ _____
   _____
   _____

🎨 Part 4: Design Challenge

Instructions: Switch to the "Playground" tab and design your own track.

1. Create a track where the skater reaches maximum speed without flying off.
   - Sketch your track below or describe its shape:
     ✎ _____
     _____
2. What design features helped maintain energy efficiency?
   ✎
   _____
   _____

🧠 Reflection

1. In your own words, explain how energy is conserved in the simulation.
   ✎
   _____
   _____

_____

_____

2. Why is this concept important in real-world engineering applications?
   ✎